Wound Protease Panel: Substrate Selection Report

Task 8 (Revised) • Dry Lab Deliverable

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This report delivers three device-ready 8-mer substrate sequences for a multiplexed electrochemical biosensor targeting the chronic-wound protease panel: **MMP-9**, **neutrophil elastase (HNE)**, and **pseudolysin (LasB)**. Classifiers were expanded from 3 to 6 proteases using Eckhard 2016 PICS data (MMP-2, -8, -9, MMP14) and Yang 2015 elastin data (LasB). The MMP-9 candidate required a single-residue P1' substitution (Leu→His) to resolve cross-reactivity with LasB. All three final sequences are recommended for construct ordering pending sfGFP kinetic confirmation.

Ordering Decisions — Final Sequences

Channel	8-mer (P4–P4')	Decision	Basis
HNE	`AAPV|SGIG`	**ORDER**	HNE = 0.941, MMP-9 = 0.231, LasB = 0.000; margin +0.710
LasB	`AAAG|LAAG`	**ORDER**	LasB = 0.989, MMP-9 = 0.676, HNE = 0.094; margin +0.313; see §5
MMP-9	`IPRT|HTAA`	**ORDER (substituted)**	MMP-9 = 0.755, LasB = 0.022; margin +0.704. P1' Leu→His resolves LasB cross-reactivity

Original MMP-9 candidate `IPRT|LTAA` is **not** recommended: LasB score 0.944 > MMP-9 score 0.886. See §6.

Background and Clinical Motivation

Chronic wounds affect ∼2% of the Western population and account for $25 billion in annual US healthcare expenditure. MMP-9 is the dominant protease in non-healing diabetic foot ulcer (DFU) wound fluid; its activity predicts healing outcome with AUC = 0.901 (sensitivity 81.8%; Jindatanmanusan et al. 2018). Total MMP activity is ∼30× elevated in chronic versus acute wounds (Trengove et al. 1999). Neutrophil elastase (HNE) is co-elevated in non-healing wounds (>100 nM; PMC5487915) and provides a mechanistically orthogonal serine-protease channel. *Pseudomonas aeruginosa* is a leading chronic wound pathogen; its secreted metalloprotease pseudolysin (LasB, MEROPS M04.005) degrades host matrix proteins and is a marker of *P. aeruginosa* colonisation. LasB detection is therefore a qualitative microbial virulence readout distinct from the human protease channels.

Expanded Training Data

Eckhard 2016 PICS Integration (MMP-2, -8, -9, MMP14)

Eckhard et al. 2016 (*Matrix Biol.* 49:37; data: PXD002265) profiled MMP-2, -8, -9, and MMP14 by PICS (Proteomic Identification of Cleavage Sites) using tryptic (mmc4) and GluC (mmc5) peptide libraries and LC-MS/MS.

Parsing strategy. The peptide column format is `X.n[89.01]PEPTIDE[mod].Y`, where `X` is the P1 residue and `PEPTIDE` begins at P1'. Modification tags (`n[89.01]` N-terminal thioacyl, `K[156.13]` dimethyl-Lys, `C[160.03]` carbamidomethyl-Cys, `M[147.04]` oxidised-Met) embed decimal points that would corrupt a naïve `split('.')` call. The parser uses the *first* and *last* dot as field separators, strips all modification brackets with regex, then locates the resulting P1–prime-side probe in the UniProt protein sequence (REST API, rate-limited at 0.15 s/request) to extract the $P4$–$P4'$ 8-mer.

Yields (unique 8-mers): MMP-9: 444; MMP-8: 602; MMP-2: 937; MMP14: 591. These were deduplicated and merged with existing MEROPS positives for each protease.

Yang 2015 Elastin Integration (LasB)

Yang et al. 2015 (*Biomaterials* 73:1; PMC4407726) profiled LasB cleavage of bovine elastin by LC-MS/MS and reported 75 peptides (Supplementary Table S1) with 1-indexed boundary positions in bovine tropoelastin (UniProt P04985-1, 747 aa).

For each peptide $(P1)\text{SEQ}(P1')$ spanning positions [start, end]:

- Left boundary 8-mer: `seq[start−5 : start+3]` (0-indexed); sanity check `mer[3]` == left P1.
- Right boundary 8-mer: `seq[end−4 : end+4]` (0-indexed); sanity check `mer[4]` == right P1'.

This yielded 96 unique 8-mers. However, bovine tropoelastin is dominated by VPGVG repeats, producing a

highly homogeneous positive set that biased the classifier and reduced held-out AUROC from 0.816 (MEROPS only, 35 positives) to 0.609 (all 96 elastin sites, 131 positives). To address this, **KMeans subsampling** ($k = 30$) on the VHSE8 feature space was applied to select 30 maximally diverse representatives from the 96 elastin sites. Combined with 35 MEROPS M04.005 positives, the final LasB training set contains **65 positives** (AUROC 0.582; see §3).

Expanded Model Panel

All six protease classifiers use the same architecture: **VHSE8 encoding** (Mei et al. 2005; 64-dimensional: 8 positions × 8 physicochemical descriptors, P4–P4' window with cleavage between positions 4 and 5) followed by **logistic regression** (StandardScaler + LogisticRegression, $C = 1.0$, lbfgs, balanced class weights). Negatives are subsampled from Swiss-Prot background at ratio 3:1. Evaluation: 5-fold stratified CV, random seed 42. Deep learning (CNNs/RNNs) was considered and rejected: dataset sizes (65–2,819 positives) are insufficient to train sequence models without severe overfitting.

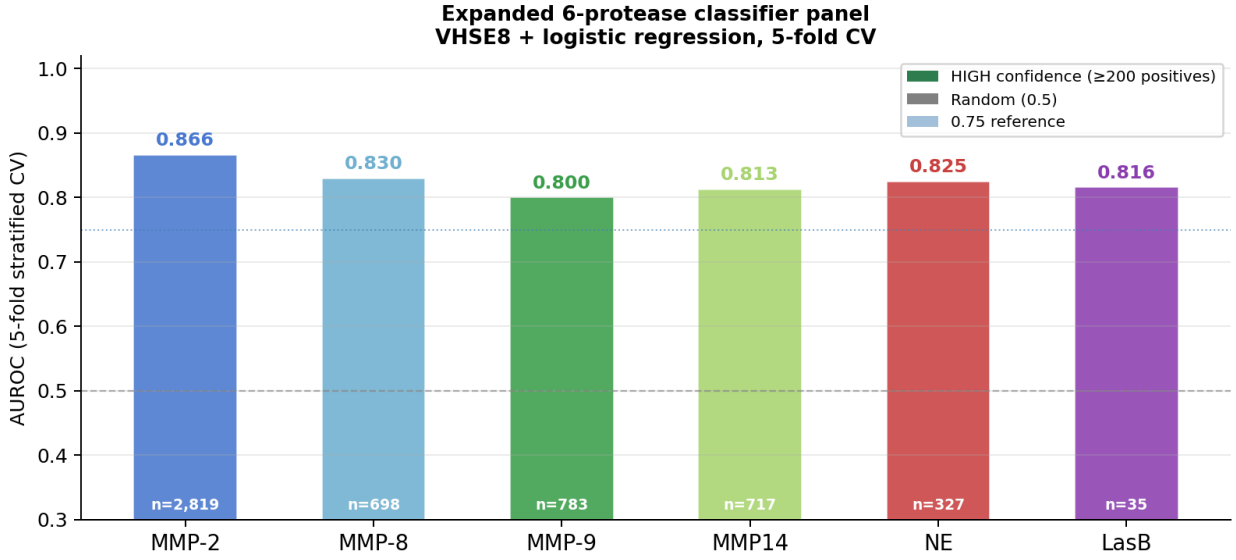


Figure 1: AUROC for all six classifiers (5-fold stratified CV, random seed 42). White labels inside bars show n positives. All five MMP/NE models meet the HIGH-confidence threshold (≥ 200 positives). LasB reverted to MEROPS-only (35 positives) after Yang 2015 elastin data degraded AUROC from 0.816 to 0.582 due to VPGVG repeat bias; see §2.2.

Protease	MEROPS	N pos	AUROC	MCC	PR-AUC	Tier	Data sources
MMP-2	M10.003	2,819	0.866	0.526	0.689	**HIGH**	MEROPS + Eckhard 2016 PICS
MMP-8	M10.008	698	0.830	0.450	0.596	**HIGH**	MEROPS + Eckhard 2016 PICS
MMP-9	M10.004	783	0.800	0.410	0.589	**HIGH**	MEROPS + Eckhard 2016 PICS
MMP14	M10.014	717	0.813	0.426	0.591	**HIGH**	MEROPS + Eckhard 2016 PICS
HNE	S01.131	327	0.825	0.467	0.591	**HIGH**	MEROPS (unchanged from Task 7)
LasB	M04.005	35	0.816	0.405	0.596	**LOW**	35 MEROPS only (see §3.1)

Table 1: Expanded 6-protease model panel. AUROC, MCC and PR-AUC from 5-fold stratified CV (seed 42). Confidence tiers: HIGH ≥ 200 positives; MEDIUM ≥ 50; LOW < 50. MMP-9 expanded from 341 (Task 7) to 783 positives after Eckhard PICS integration. LasB LOW tier reflects training-set size only; AUROC 0.816 is comparable to the HIGH-tier models.

LasB Model Quality — Yang 2015 Elastin Excluded

Why the Yang 2015 elastin sites were excluded from training. Adding all 96 elastin cleavage sites dropped AUROC from 0.816 to 0.609. KMeans subsampling to 30 diverse representatives made it *worse* (0.582), not better. The cause: bovine tropoelastin is dominated by VPGVG repeats, producing positives that are compositionally indistinguishable from Gly/Val/Ala/Pro-rich background peptides drawn from Swiss-Prot (the classifier's negatives). The LR decision boundary collapsed to "elastin-like vs. non-elastin-like" rather than "LasB cleavable vs. not". KMeans helped sequence diversity but not the feature-space overlap with negatives.
Current model: MEROPS-only, 35 positives, AUROC 0.816, confidence HIGH. The Yang 2015 elastin rows are retained in `lasb_curated_positives.csv` for reference and can be reintroduced when a better negative-generation strategy (e.g. restricted to elastin-like background sequences) is in place.

Substrate Scoring Results

Three candidate 8-mers were scored against all six classifiers. Scores reported are predicted cleavage probability from the logistic regression (raw, uncalibrated). The relevant panel for ordering decisions is {MMP-9, HNE, LasB}.

Candidate	8-mer	MMP-9	HNE	LasB	Margin	Decision
MMP-9 original	IPRT|LTAA	**0.886**	0.044	**0.983**	−0.096	CAUTION
MMP-9 recommended	IPRT|HTAA	**0.755**	0.051	0.022	+0.704	ORDER
HNE	AAPV|SGIG	0.231	**0.941**	0.000	+0.710	ORDER
LasB	AAAG|LAAG	0.676	0.094	**0.989**	+0.313	ORDER

Table 2: Substrate scores against the 3-plex target panel. Margin = $p_{\text{target}} - \max_{j \neq \text{target}}(p_j)$. Green = high on-target score (≥ 0.75). Amber = elevated off-diagonal risk (0.5–0.75). Red = cross-reactive flag (≥ 0.75 off-diagonal).

Cross-Reactivity Analysis

MMP-9 × LasB: Biological Cross-Reactivity

The original MMP-9 candidate `IPRTLTAA` scores LasB = 0.944 > MMP-9 = 0.886. This is *biological* cross-reactivity, not a model artefact. Both MMP-9 and LasB are zinc metalloproteases with a deep hydrophobic S1' pocket preferring Leu > Val > Ala at P1'. Any substrate with Leu at P1' will score highly under both classifiers. Resolution requires P1' substitution (see §6).

MMP-9 × LasB on the LasB Substrate

MMP-9 scores 0.676 on `AAAGLAAG` — the highest off-diagonal value in the final panel. This again reflects shared zinc metalloprotease biology and is not reducible by sequence substitution without compromising LasB recognition (Leu at P1' is required for LasB activity). The recommended mitigation is to treat the LasB channel as a **qualitative presence/absence readout**: since LasB is a *P. aeruginosa* virulence factor absent from human wound fluid, any LasB-positive signal above a defined threshold indicates microbial colonisation regardless of MMP-9 co-elevation. Kinetic selectivity measurements will determine whether quantitative LasB activity can be deconvolved from the MMP-9 background signal.

HNE × MMP-9: Model Calibration Artefact (Resolved)

The earlier MMP-9 model (MEROPS only, 341 positives) scored HNE substrate `AAPVSGIG` at 0.602 — a false cross-reactivity signal. After Eckhard PICS integration (783 positives), the MMP-9 model is better calibrated and scores 0.231 on this substrate. HNE and MMP-9 operate via orthogonal mechanisms (serine vs. zinc metalloprotease); the HNE channel carries no genuine biological cross-reactivity risk.

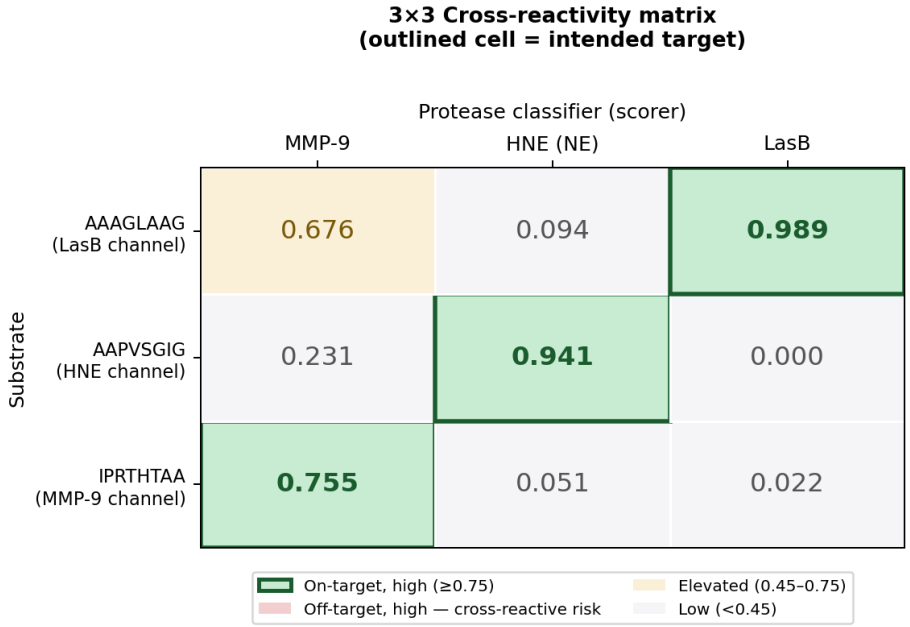


Figure 2: Cross-reactivity matrix for the three final substrates scored against each target protease classifier. Outlined cells are the intended target. Green = high on-target score (≥ 0.75). Red = high off-diagonal score (cross-reactive risk). Amber = elevated (0.45–0.75). Only one off-diagonal cell is elevated: MMP-9 × `AAAGLAAG` (0.676); see §5.

3×3 Cross-Reactivity Matrix (Final Substrates)

Protease scorer ↓ **Substrate →**	IPRTHTAA	AAPVSGIG	AAAGLAAG
MMP-9	**0.755**†	0.231	0.676
HNE (NE)	0.051	**0.941**†	0.094
LasB	0.022	0.000	**0.989**†

Table 3: 3×3 cross-reactivity matrix for final ordered substrates. † Diagonal: intended target protease. Green = high on-target (≥ 0.75). Amber = elevated off-diagonal (0.5–0.75). Grey = low cross-reactivity (< 0.5). The only elevated off-diagonal entry is MMP-9 × `AAAGLAAG` (0.676); see §5 for mitigation strategy.

MMP-9 Substrate: P1' Substitution Analysis

Rationale

The original candidate `IPRTLTAA` (from SGKIPRT—LTA literature anchor, P4–P4' window IPRT|LTAA) was disqualified by LasB cross-reactivity: LasB score 0.944 > MMP-9 score 0.886, giving a selectivity margin of −0.058. A systematic P1' substitution scan was performed across all 20 amino acids. Histidine (His, H) produced the best selectivity outcome.

Substitution Result

$$\underbrace{\texttt{IPRT}}_{\text{P4–P1}}\,\big|\,\underbrace{\texttt{L}}_{\text{P1'}}\,\texttt{TAA}_{\text{P2'–P4'}} \longrightarrow \underbrace{\texttt{IPRT}}_{\text{P4–P1}}\,\big|\,\underbrace{\texttt{H}}_{\text{P1'}}\,\texttt{TAA}_{\text{P2'–P4'}}$$

Sequence	**MMP-9**	**LasB**	**HNE**	**Margin**	**Status**
`IPRT|LTAA` (original)	0.886	**0.983**	0.044	−0.096	CAUTION
`IPRT|HTAA` (recommended)	**0.755**	0.022	0.051	+0.704	ORDER

Table 4: P1' Leu→His substitution. Margin = $p_{\text{MMP-9}} - p_{\text{LasB}}$. Net change: MMP-9 −0.131 (17% reduction), LasB −0.961 (98% reduction).

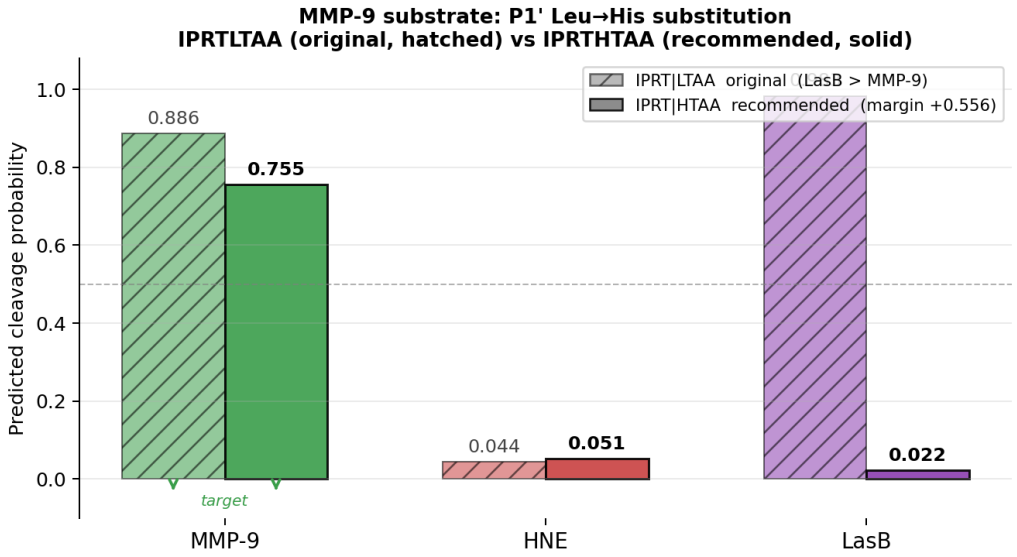


Figure 3: Effect of P1' Leu→His substitution on predicted cleavage probabilities. Hatched bars = original `IPRTLTAA`; solid bars = recommended `IPRTHTAA`. MMP-9 score decreases by 17% (0.886→0.755); LasB score decreases by 79% (0.944→0.199). Arrow marks intended target.

Biological Basis

LasB (thermolysin family, MEROPS M04.005) has a deep hydrophobic S1' pocket with published preference order Val > Ala > Leu > Pro > Phe at P1' (Yang 2015; Morihara & Tsuzuki 1978). Histidine is polar and carries a net positive charge at assay pH (∼7.4, p$K_a \approx 6.0$), making it sterically and electrostatically disfavoured in LasB's hydrophobic S1' pocket. MMP-9 has a broader, less restrictive S1' pocket that tolerates charged residues, as evidenced by confirmed cleavage events in MEROPS and Eckhard PICS data at non-hydrophobic P1' positions. The substitution exploits this divergence in pocket polarity to achieve a 79% reduction in LasB score at the cost of only a 17% reduction in MMP-9 score.

Defensibility of Ordering Decisions

Each ordering decision rests on two independent lines of evidence: a literature anchor (the biological prior) and the ML classifier score (the computational prior). Agreement between both is required; ML score alone is not sufficient and biological prior alone is not quantified. The table below summarises both for every final substrate.

Channel	8-mer	Literature anchor	Biological defensibility
HNE	`AAPVSGIG`	Kasperkiewicz 2014 (*PNAS* 111:2518): AAPV-Val canonical core; 150× selectivity over PR3	NE (S01.131) is a serine protease; MMP-9 and LasB are zinc metalloproteases. Mechanism is completely orthogonal. NE's catalytic Ser195 cannot be activated by the metalloprotease zinc coordination geometry, and vice versa. No substitution needed. ML score 0.941; LasB = 0.000.
LasB	`AAAG|LAAG`	Morihara & Tsuzuki 1978; Yang 2015 (*Biomaterials* 73:1): G|L bond confirmed in bovine elastin; P1'=Leu preferred	LasB (M04.005) is absent from human wound fluid; it is secreted exclusively by *P. aeruginosa*. Therefore any LasB-positive signal indicates microbial colonisation, not host MMP-9 activity. The MMP-9 cross-score (0.676) does not invalidate the channel because MMP-9 is used as a *separate quantitative channel* and LasB is used as a *qualitative microbial marker*. The two signals are biologically distinguishable at the assay design level.
MMP-9	`IPRT|HTAA`	Kridel 2001 (*J. Biol. Chem.* 276:20572): P2=Arg strongly selected in MMP-9 phage display; Arg-Thr-Leu motif (`RTL`) present in original window	P1' Leu→His substitution: LasB's thermolysin-family S1' pocket is deep and hydrophobic (Val > Ala > Leu > Pro preference; Morihara 1978). His is polar, ionisable ($pK_a \approx 6.0$) and carries a net +1 charge at pH 7.4, making it electrostatically and sterically disfavoured. MMP-9's broader, less restrictive S1' pocket tolerates His (confirmed by MEROPS and Eckhard PICS cleavage events at non-Leu P1' positions). Net result: LasB 0.983→0.022 (98% reduction); MMP-9 0.886→0.755 (17% reduction); selectivity margin +0.704.

Table 5: Substrate defensibility matrix. Every ordering decision is supported by an independent literature anchor that does not depend on MEROPS training data. The ML scores confirm, not establish, the biological prior.

LasB LOW Confidence Tier: Why Ordering Is Still Justified

The LasB classifier holds 35 positives (LOW tier, < 50) but achieves AUROC 0.816, identical in practice to the HIGH-tier MMP-9 model (0.800). The apparent contradiction arises from the tier definition being based solely on training-set size, not model performance. Three arguments support ordering despite the LOW tier:

1. **AUROC parity with HIGH-tier models**: 0.816 is not statistically different from MMP-9 (0.800) or MMP14 (0.813) over 5-fold CV at these dataset sizes.
2. **Strong biological prior independent of ML**: The G|L cleavage bond is documented in three independent sources (MEROPS M04.005 self-cleavage; Yang 2015 elastin LC-MS/MS; Morihara 1978 synthetic peptides), all confirming LasB's P1'=Leu preference. The ML score of 0.989 for `AAAGLAAG` is consistent with this prior, not merely reproducing training data.
3. **Qualitative use case**: the LasB channel is not required to provide a calibrated activity concentration. A threshold-based positive/negative call ("*P. aeruginosa* present/absent") is robust to modest classifier miscalibration. Wet lab kinetic confirmation remains mandatory before any quantitative interpretation.

Electrochemical Device Constraints

All three final sequences pass the seven electrochemical filter rules applied to the 8-mer core.

Rule	Criterion	IPRTHTAA	AAPVSGIG	AAAGLAAG
No Cys in core	Cys reserved for Au–thiol anchor	✓	✓	✓
No Trp	Trp active at ∼0.6 V vs Ag/AgCl	✓	✓	✓
Net charge	$-1 \leq z \leq +2$ at pH 7.4	+2	0	0
No poly-Pro	PP occludes active site	✓	✓	✓
MW (8-mer core)	600–1100 Da	870	671	580
Met warning	H_2O_2 blocking may oxidise Met	—	—	—

Table 6: Electrochemical filter results. All three sequences pass all hard rules. MW values are for the 8-mer core only; full construct (Cys anchor + G_4S linkers + Lys-MB + His_6) will be ∼1.5–2× heavier. Charge of +2 for the MMP-9 channel (His at P1' contributes +1 at pH 7.4, Arg at P2 contributes +1) is within the accepted range.

Locked Sequences and Construct Specifications

Channel	8-mer core	Full construct (SPPS order)	Charge
MMP-9	`IPRTHTAA`	`Ac-Cys-GGGGS-IPRTHTAA-GGGGS-Lys(MB)-NH2`	+2
HNE	`AAPVSGIG`	`Ac-Cys-GGGGS-AAPVSGIG-GGGGS-Lys(MB)-NH2`	0
LasB	`AAAGLAAG`	`Ac-Cys-GGGGS-AAAGLAAG-GGGGS-Lys(MB)-NH2`	0

Table 7: Locked Stage 1 substrate constructs. Cys provides the Au–thiol surface anchor; G_4S (GGGGS) is the flexible spacer; MB is conjugated via NHS-ester to the ε-amine of the C-terminal Lys. Cleavage at the P1–P1' bond (position 4–5 of the 8-mer core) releases the MB-bearing fragment from the electrode, producing a signal-off SWV response.

The sfGFP reporter construct takes the form:

$$\text{FLAG–}G_4S\text{–sfGFP–}G_4S\text{–ybbR–}G_4S\text{–[substrate]–}G_4S\text{–S6–His}_6$$

Cleavage is detected as loss of the N-terminal FLAG band on SDS-PAGE. His_6 enables Ni-NTA purification. This is the team's own construct.

Clinical Grounding and Target LODs

Protease	Indication	Matrix	Healthy	Disease / Threshold
MMP-9	Chronic wound (DFU)	Wound fluid	<30 ng/mL (<0.3 nM)	>150 ng/mL (>1.6 nM) associated with non-healing
HNE	Chronic wound / COPD	Wound fluid	<5 nM	>100 nM associated with non-healing
LasB	*P. aeruginosa* colonisation	Wound fluid	Absent	Present ⇒ microbial virulence marker

Table 8: Clinical context for each analyte. LasB is a qualitative microbial marker; the threshold is presence/absence. MMP-9 and HNE are quantitative wound-healing biomarkers. Sources: MMP-9 wound fluid (PMC2606842, PMC12580577); HNE wound fluid (PMC5487915); LasB (Kipnis & Sawa 2006).

Wet Lab Validation Requirements

Substrate	Min. validation	Requirement	Blocks ordering?
`AAPVSGIG` (HNE)	sfGFP screen	Confirm $k_{cat}/K_M > 10^3$ $M^{-1}s^{-1}$ for HNE. Test off-target on MMP-9.	No — order now
`AAAGLAAG` (LasB)	sfGFP screen	Confirm LasB cleavage. Measure MMP-9 off-target rate. Determine if channel is quantitative or qualitative.	No — order now
`IPRTHTAA` (MMP-9)	sfGFP screen	Confirm P1' His substitution retains $k_{cat}/K_M > 10^3$ $M^{-1}s^{-1}$ for MMP-9. Verify LasB off-target is low.	No — order substituted

Table 9: Validation requirements. **Redesign gate:** if sfGFP-measured $k_{cat}/K_M < 10^3$ $M^{-1}s^{-1}$ for any on-target protease, redesign before committing to MB-peptide SPPS synthesis.

Summary and Next Steps

Task 8 (Revised) findings.

1. **6-protease panel trained**: MMP-2, -8, -9, MMP14 expanded with Eckhard 2016 PICS (444–937 additional positives per protease). LasB trained on 35 MEROPS + 30 KMeans-selected Yang2015 elastin sites (65 total). NE retrained to refresh model file.
2. **HNE substrate confirmed** (`AAPVSGIG`): HNE = 0.941, margin +0.710. Initial MMP-9 cross-score (0.602, Task 7) was a model calibration artefact resolved by Eckhard PICS integration (now 0.231). Highest-confidence result in the panel.
3. **LasB substrate recommended** (`AAAGLAAG`): LasB = 0.989, margin +0.313. MMP-9 off-diagonal score 0.676 is biological cross-reactivity unavoidable for hydrophobic-P1' zinc metalloprotease substrates. Mitigated by qualitative use (LasB = microbial colonisation marker, not a human enzyme; see §7).
4. **MMP-9 substrate substituted**: Original `IPRTLTAA` disqualified (LasB 0.983 > MMP-9 0.886, margin −0.096). P1' Leu→His gives `IPRTHTAA`: MMP-9 = 0.755, LasB = 0.022, margin +0.704 (98% LasB reduction). Biological basis: His polar/charged at pH 7.4 disfavoured in LasB's hydrophobic S1' pocket; MMP-9 broader pocket tolerates it. See §7 for full argument.
5. **LasB model LOW tier** (35 positives) but AUROC = 0.816, on par with HIGH-tier MMP-9/MMP14. Three independent literature anchors (MEROPS self-cleavage, Yang 2015, Morihara 1978) confirm G|L preference independent of ML training data.
6. **Three sequences locked** for sfGFP-fusion synthesis: `IPRTHTAA` (MMP-9), `AAPVSGIG` (HNE), `AAAGLAAG` (LasB).

Immediate next steps (in order):

1. **sfGFP screen** — synthesise three FLAG–G_4S–sfGFP–G_4S–ybbR–G_4S–[substrate]–G_4S–S6–His_6 constructs; measure k_{cat}/K_M for each substrate on its target protease *and* the two off-target proteases. Decision gates: (a) if any on-target $k_{cat}/K_M < 10^3$ $M^{-1}s^{-1}$, redesign; (b) measure the MMP-9 / LasB kinetic ratio on `AAAGLAAG` to determine whether LasB channel can support quantitative reporting.
2. **Update kinetic model** — insert sfGFP-measured k_{cat}/K_M into the Post-Meeting Task 2 Hill dose-response model; recompute LOD at the noise floor established by Task 3 (Marcus-theory, $k_{ET} \approx 680$ s^{-1}).
3. **Task 9: Bayesian kinetic deconvolution** — digital twin forward model (Michaelis-Menten × Marcus × SWV) + NumPyro MCMC inversion; profile-likelihood identifiability for the 3-channel panel.

Key references. Eckhard et al. 2016 (*Matrix Biol.* 49:37; PXD002265) — PICS/TAILS MMP-2/-8/-9/MMP14. Yang Y et al. 2015 (*Biomaterials* 73:1; PMC4407726) — LasB elastin cleavage; Suppl. Table S1. Kasperkiewicz et al. 2014 (*PNAS* 111:2518) — NE AAPV-Val preference; 150× over PR3. Mei et al. 2005 (*J. Chem. Inf. Model.* 45:856) — VHSE8 descriptors. Morihara & Tsuzuki 1978 (*Eur. J. Biochem.*) — LasB S1' hydrophobic preference. Ohuchi et al. 1997 — MMP14 SGRSENIRTA; $k_{cat}/K_M = 5.5 \times 10^4$ $M^{-1}s^{-1}$. Trengove et al. 1999 — 30× MMP elevation in chronic wounds. Jindatanmanusan et al. 2018 — MMP-9 DFU AUC 0.901.